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A PALEOCENE PALEOMAGNETIC POLE FOR NORTH AMERICA FROM ALKALIC INTRUSIONS, CENTRAL MONTANA

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Abstract. The apparent polar wandering path from Late Cretaceous through early Tertiary time is not well defined. We have collected 33 paleomagnetic sites from the Paleocene Moccasin, Judith, and Little Rocky Mountains and from these a North American Paleocene paleomagnetic pole position has been determined at 174.9°W long., 80.5°N lat., with a circle of 95% confidence of 5.6°. This pole location serves as a Paleocene reference pole and enhances the North American apparent polar wandering path for a critical period of its history.

Introduction

The North American pole position for the Cretaceous has been well established from studies using rocks older than 80 m.y. (McElhinney, 1973), while the pole position from at least Oligocene time is indistinguishable from the present geomagnetic pole position (Diehl et al., 1974). The pole location from Late Cretaceous through Eocene time, however, is not well defined. We report here on a new paleomagnetic pole position for the Paleocene. A well-defined Paleocene pole is particularly needed for several reasons. First, the ages of rotated blocks in the Oregon-Washington Coast Range mostly fall in the "gap" between Late Cretaceous and Oligocene time for which highly reliable North American reference poles have not been determined. Also, Coney (1972, 1978) has shown that specific orogenies may be correlated with plate interactions. For example, the Laramide Orogeny may be an expression of the stresses imposed upon a rapidly moving North American plate between 80 and 40 m.y.b.p. The establishment of a North American polar wandering curve based upon accurately determined paleomagnetic pole positions for the Paleocene and Eocene epochs can aid in plate-motion reconstructions. Prior to this study only one North American Paleocene pole had been determined (Butler and Taylor, 1978), and this may be biased by an inclination error (Barnes and Butler, 1980).

Geology and Ages of Intrusions

The Moccasin, Judith, and Little Rocky Mountains are the exposed roots of intrusive centers within the alkalic province of central Montana. Figure 1 shows the locations of these mountains.

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A brief description of the geology and ages of the intrusive structures is given below. The ages given for each lithology are from Marvin et al. (1980) and are either the result of one isotopic age determination, the average of a number of isotopic ages, or the most reasonable age as determined from isotopic dating and geological considerations.

The Moccasin Mountains consist of North and South Moccasin Mountain, each a laccolith with associated dikes and sills (Blixt, 1933; Miller, 1959; Lindsey, 1980). North Moccasin Mountain is composed of a syenite porphyry, referred to here as the North Moccasin syenite porphyry, while South Moccasin Mountain is an intrusive complex consisting of a syenite porphyry, alkali syenite porphyry, and quartz monzonite porphyry. Paleomagnetic sites were collected in the North Moccasin syenite porphyry, and quartz monzonite porphyry and syenite porphyry of South Moccasin Mountain which have been dated at 66.0, 64.2, and 63.7 m.y., respectively.

Geological mapping in the Judith Mountains was done by Wallace (1953). The Judith Mountains are an intrusive complex with numerous stocks, laccoliths, dikes, and sills of several rock types. Lithologies sampled in this area for paleomagnetic investigation include a quartz monzonite, the Judith Peak alkali granite, a rhyolite, and a trachyte. The quartz monzonite and Judith

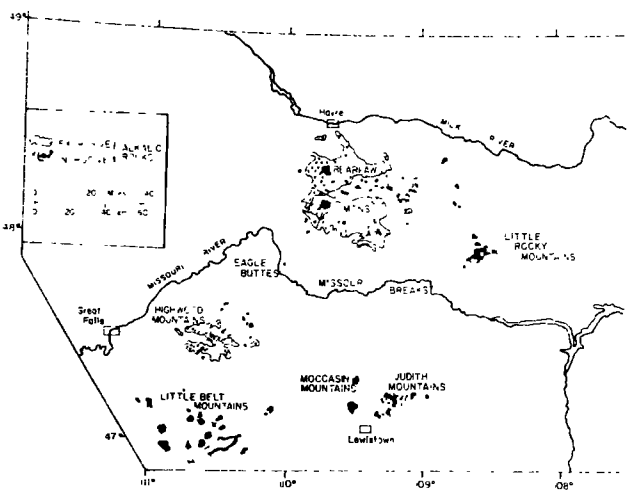


Fig. 1. Map showing the locations of the Moccasin, Judith, and Little Rocky Mountains. Redrawn from Hearn et al. (1977).

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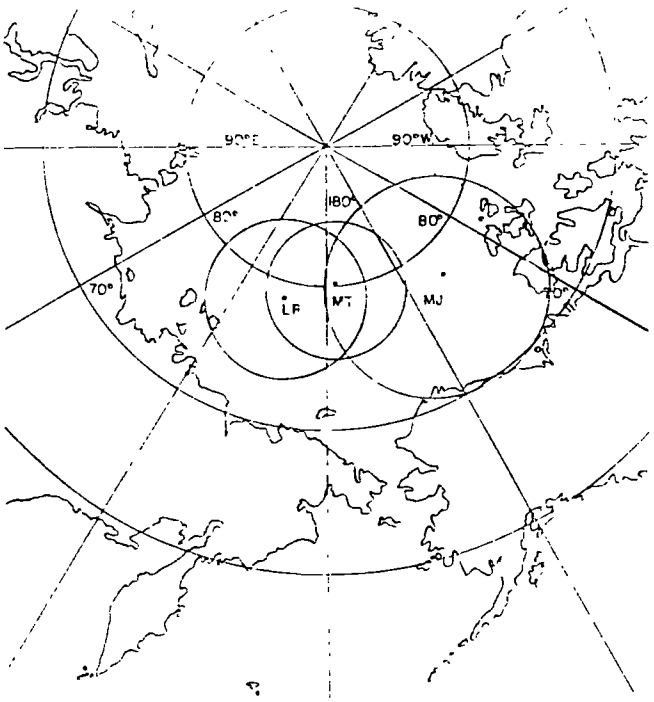


Fig. 2. Location of the Little Rockies (LR), Moccasin-Judith (MJ), and Montana (MT) poles with their respective circles of 95% confidence.

Peak alkali granite each have given a range of ages by K/Ar, fission-track, and lead-uranium-thorium dating methods, and the best estimates of their ages are 65.0 and 62.0 m.y., respectively. The rhyolite and trachyte have not been dated but Forrest (1971) considers the rhyolite to be younger than the quartz monzonite, and geological mapping by Hall (1975) shows that the trachyte post-dates the alkali syenite.

The geology of the Little Rocky Mountains was described by Brockunier (1936), Knechtel (1959), and Bailey (1974). The mountains are an intrusive complex consisting of several rock types. Laccoliths or stocks, sills, and dikes are present. Using the terminology of Brockunier, the lithologies we sampled for this study and their ages are as follows: Antone syenite porphyry, between 67.0 and 62.4 m.y.; Old Scraggy syenite porphyry, 62.5 m.y., but perhaps older; Silvertip alaskite, between 67.0 and 61.2 m.y.; Indian Peak hornblende syenite porphyry, 61.8 m.y.; and aegirine trachyte, 61.0 m.y.

Results

A total of 38 paleomagnetic sites (336 samples) were collected for this study, 14 in the contiguous Moccasin and Judith Mountains and 24 in the Little Rocky Mountains. Cores were drilled and oriented following the procedures out-

TABLE 1. Paleomagnetic data for Paleocene intrusive rocks from central Montana.

Site	$\bar{D}$	$\bar{I}$	α_{95}	K	N	λ'(°N)	ϕ'(°)
MJ 1	169.1	-72.9	7.8	35.3	9	77.0	135.4 W
MJ 3	337.2	62.0	14.5	10.2	9	73.5	154.5 E
MJ 4	30.6	82.7	11.9	19.7	7	58.9	95.1 W
MJ 5	219.5	-76.8	7.1	48.5	8	62.4	73.5 W
MJ 7	319.8	69.5	7.7	41.0	8	63.9	170.8 W
MJ 8	353.6	71.4	2.2	508.0	8	80.3	131.2 W
MJ 9	350.1	65.5	6.8	46.9	9	83.3	191.4 W
MJ 10	332.2	67.6	2.3	408.9	9	71.6	178.5 W
MJ 11	6.9	70.7	3.4	192.9	9	81.0	83.0 W
MJ 15	333.6	80.7	2.0	465.5	10	62.4	126.7 W
MJ 16	349.7	63.4	3.9	142.9	9	82.5	147.0 E
LR 2	340.0	49.5	5.4	74.0	9	66.7	119.8 E
LR 3	332.4	52.1	11.7	28.6	5	64.3	135.3 E
LR 4	330.8	62.8	5.4	73.8	9	69.5	198.2 W
LR 5	282.5	69.7	4.1	103.9	11	43.1	161.3 W
LR 6	347.8	58.2	4.8	92.9	9	77.4	120.3 E
LR 7	357.1	69.1	3.2	208.1	9	84.1	125.3 W
LR 8	343.7	61.1	8.1	33.0	9	77.2	140.7 E
LR 9	7.4	69.9	5.2	91.3	8	82.5	73.1 W
LR 10	0.9	46.5	6.2	64.5	8	69.9	69.0 E
LR 11	22.4	73.1	5.5	81.5	8	73.0	66.2 W
LR 12	12.7	68.2	6.1	74.9	7	81.1	45.7 W
LR 13	316.1	52.5	4.9	118.8	7	54.1	153.4 E
LR 14	323.9	52.7	5.5	71.8	9	59.3	146.2 E
LR 16	359.2	74.3	4.6	83.6	11	77.3	110.3 W
LR 18	324.3	59.4	5.4	66.4	10	63.5	157.9 E
LR 19	325.7	61.3	5.5	80.6	8	65.5	161.4 E
LR 21	339.1	58.3	5.4	94.4	7	72.5	138.5 E
LR 22	32.5	58.7	12.1	16.6	8	65.2	10.6 W
LR 23	347.8	59.4	4.7	96.4	9	78.3	124.4 E
LR 24	357.6	66.3	1.9	570.8	9	88.2	168.8 W
LR 25	348.2	71.9	5.9	62.2	9	78.6	142.8 W
LR 26	4.9	74.0	6.8	42.0	10	77.4	97.5 W

$\bar{D}$, $\bar{I}$ are mean site directions; α_{95} is the radius of the circle of 95% confidence about the mean; K is the precision parameter (Fisher, 1953); N is the number of specimens for each site; λ', ϕ' are latitude and longitude of site VGP.

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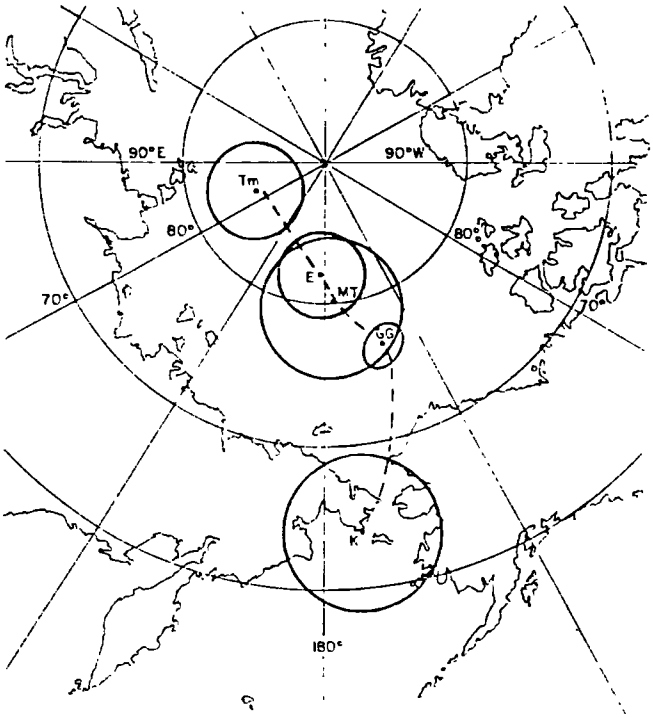


Fig. 3. Location of the Cretaceous (K), Montana (MT), Gringo Gulch (GG), Eocene (E), and mid-Tertiary (Tm) poles with their respective circles or ovals of 95% confidence.

lined by Doell and Cox (1965). Whenever possible cores were oriented using a sun compass to eliminate the effects of local magnetic fields. Each site consisted of five to eleven samples. Each sample was cut into standard-sized specimens and their natural remanent magnetism (NRM) measured on a Schonstedt Model SSM 1-A spinner magnetometer. Two or more pilot specimens from each site were then step demagnetized by alternating field (a.f.) demagnetization on a Schonstedt Model GSD-5 AC Tumbling-Specimen Demagnetizer in fields of 50, 100, 150, 200, 300, 400, and 500 oe. Directions and intensities for pilot specimens were plotted after each demagnetization step. The demagnetization value at which directions remained stable was chosen as optimum and the entire site demagnetized at that value. Demagnetization levels between 200 and 400 oe were usually found to be best, although lower or higher values were occasionally used.

Three sites were rejected because their circles of 95% confidence (α_{95}) exceeded 15°. These sites had abnormally high intensities and random directions, suggesting remagnetization by lightning. One site in an extremely weathered outcrop of quartz monzonite porphyry had NRM intensities too weak to be measured and also was rejected. Finally, one site from the Little Rockies had a mean site inclination of less than 25°, far less than any other site (Jacobson, 1980, Appendix). We surmize that this site is affected by a local tectonic rotation or became magnetized during a polarity transition or excursion of the geomagnetic field. Consequently this site was also rejected. This left a total of 33 sites (284 samples) from which to determine the Paleocene pole.

From each mean site direction a virtual geomagnetic pole (VGP) was calculated. MJ site VGPs were combined to yield a MJ pole position at 135.7°W long., 77.3°N lat., with an α_{95} of 8.1° and a precision parameter (K) of 26.8. Combined LR site VGPs locate an LR pole at 198.7°W long., 79.9°N lat., with an α_{95} of 7.0° and a K of 17.9. As these two circles of confidence overlap, all mean site VGPs were combined to yield a North American Paleocene pole at 174.9°W long., 80.5°N lat., with an α_{95} of 5.6° and a K of 18.8. This pole, which we refer to here as the Montana pole (MT), the MJ pole, and the LR pole are shown in Figure 2 with their respective α_{95}s. Table 1 lists the pertinent data for each site.

We believe that each VGP used in this study represents an independent measurement of the North American apparent pole position during the Paleocene. The apparent age-range sampled, the scatter in between-site directions and the presence of both normal and reversely magnetized sites show that secular variation has been adequately sampled and averaged into our Paleocene pole.

Discussion

Figure 3 shows the Cretaceous pole position (K) from McElhinney (1973), a Paleocene pole determined by Barnes and Butler (1980) from the Gringo Gulch Volcanics (GG), an Eocene pole (E) from Diehl et al. (1980), and a mid-Tertiary pole (Tm) from Beck and Burr (1979). All poles are shown with their circles or ovals of 95% confidence. Figure 3 illustrates these important points: (1) Our Paleocene pole and the Paleocene pole determined by Barnes and Butler (1980) agree quite well. This strongly suggests that the North American Paleocene pole position has been accurately located. (2) The Eocene pole of Diehl et al. (1980) is indistinguishable from our Paleocene pole and is significantly different from the present pole location and the mid-Tertiary pole. (3) A smooth curve can be drawn from the Cretaceous pole through the Paleocene pole, the Eocene pole, to the mid-Tertiary pole. This apparent polar wandering path agrees well with the one determined by Irving (1977) using pole positions calculated from 40 m.y. running averages. (4) The Paleocene and Cretaceous poles are significantly different. Thus, rapid apparent polar wandering appears to have begun in latest Cretaceous time (Barnes and Butler, 1980). As Coney (1972, 1978) has suggested, this period of Earth history coincides with the beginning of the Laramide Orogeny.

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